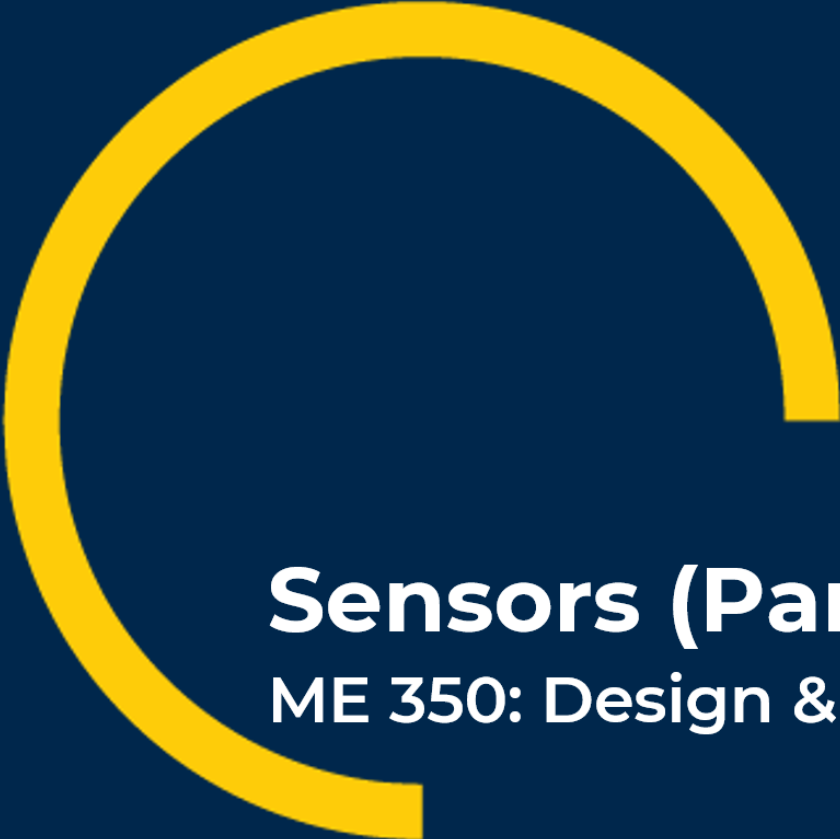




Sensors (Part 1)

ME 350: Design & Manufacturing II

Jeffrey Koller

Some slides adopted from Mike Umbriac, David Remy, Chinedum Okwudire, & Shorya Awtar

Announcements



- HW05 will now be due on Friday (10/24) at 1:30 PM
- We have created a [new google form](#) for students to fill out for reimbursements. It will just make it easier for them and ensure we receive the correct information. Please put ME 350 for the shortcode information.

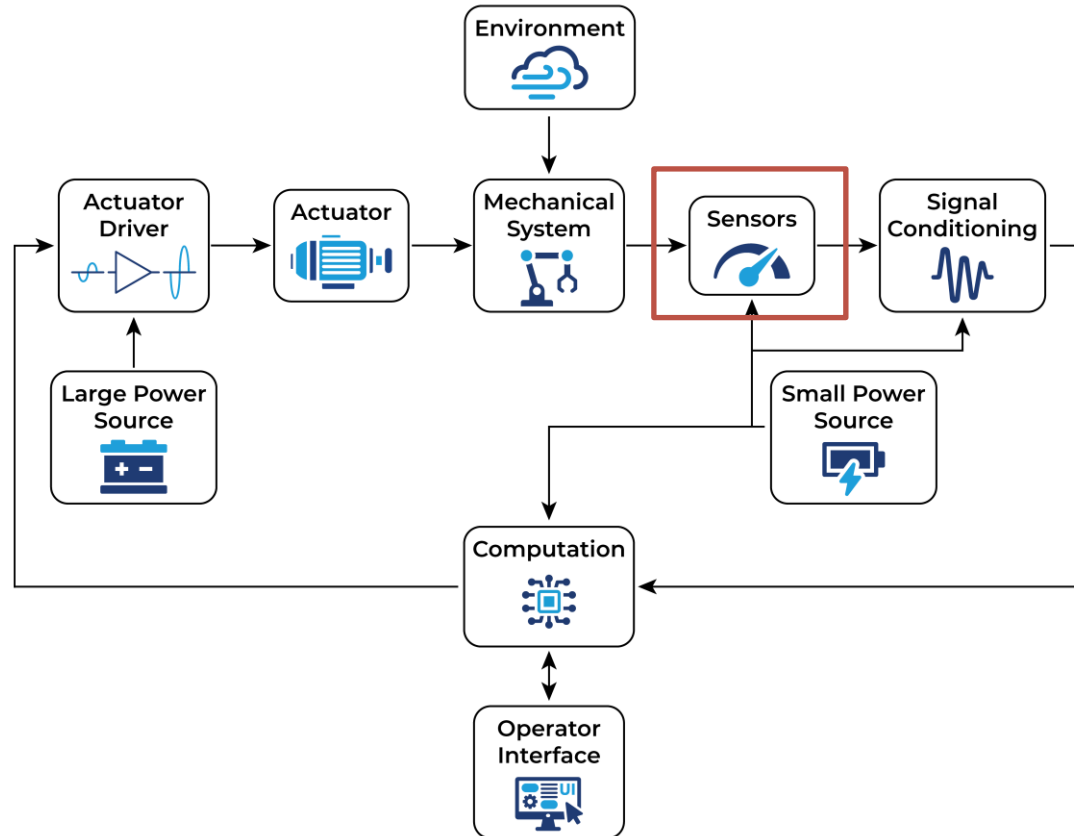
Today's Learning Objectives



By the end of today's class you should...

- Be able to answer the question: "What is a sensor?"
- Describe the difference between **digital** and **analog** sensors
- Understand how an optical encoder works and how its output signal is interpreted into something meaningful

Context for Today's Lesson



What is a Sensor?



- Take two minutes to come up with a formal definition with the person next to you.
- Definition: A **sensor** is a device that responds to a physical stimulus (i.e. heat, light, pressure, flow, etc.) and produces a **signal** for measurement or control.
- Examples of what a sensor can measure:
 - Velocity
 - Displacement
 - Position
 - Proximity (physical presence within some critical distance)
 - Liquid level
 - Pressure
 - Force

Transducer vs. Sensor

Transducer

Converts energy from one form to another



Sensor

Detects events or changes in its environment and converts that information to an **electronic** signal



Examples of Sensors



A (potentially very limited) [list on Wikipedia](#) has ~400 sensors listed separated into 17 different categories

1. Acoustic, sound, vibration
2. Automotive
3. Chemical
4. Electric current, electric potential, magnetic, radio
5. Environment, weather, moisture, humidity
6. Flow, fluid velocity
7. Ionizing radiation, subatomic particles
8. Navigation instruments
9. Position, angle, displacement, distance, speed acceleration
10. Optical, light, imaging, photon
11. Pressure
12. Force, density, level
13. Thermal, heat, temperature
14. Proximity, presence
15. Sensor technology
16. Speed sensor
17. Others

Common Sensors We Use as Mechanical Engineers



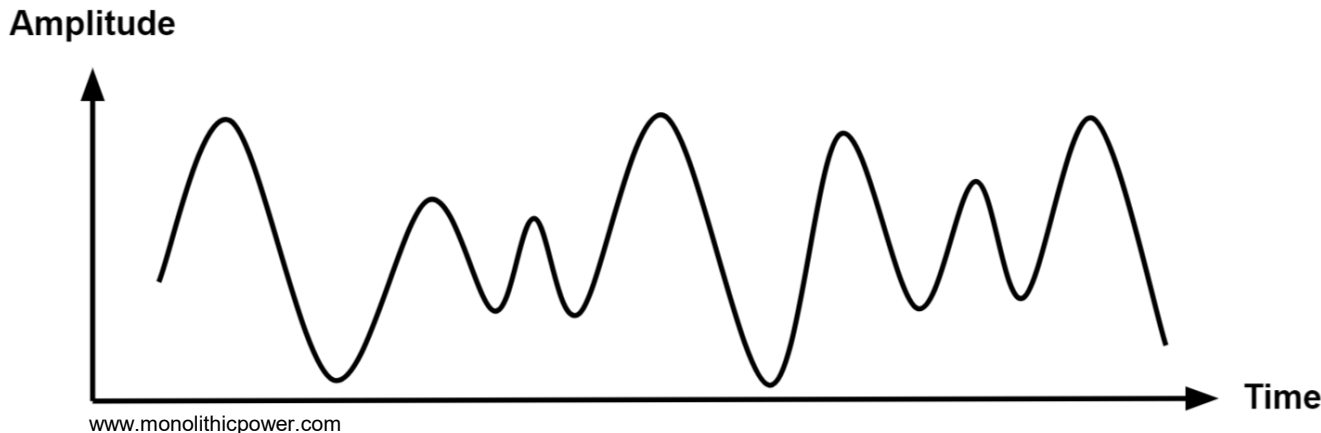
Measured Quantity	Common Sensors
Linear Displacement	Linear Potentiometer, Linear Encoder, Capacitance Displacement Sensor, Inductive Displacement Sensor, LIDAR, Linear Hall Effect Sensor, Linear Variable Differential Transducer
Angular Displacement	Rotary Encoder, Rotary Potentiometer, Angular Hall Effect Sensor, Accelerometer, Inclinator
Linear/Angular Acceleration	Accelerometers & Gyroscopes (IMUs)
Torque/Force	Load Cell (Strain Gauges)
Temperature	Thermocouples, Thermistors, Resistance Temperature Detector (RTD)
Pressure	Pressure Gauge/Sensor (Resistance, Capacitive, Piezoelectric)
Current	Shunt Resistor, Hall Effect
Proximity	Hall Effect, Optical, Inductive, Limit Switches

*List only provided for examples, no need to memorize

Analog Sensors



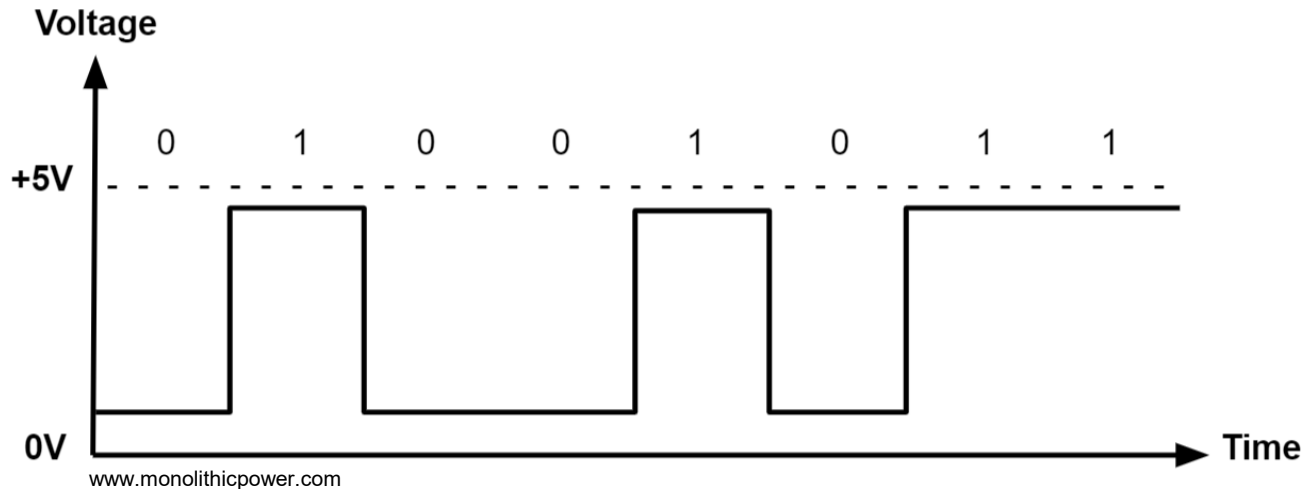
- **Analog sensors** produce an output signal that varies **continuously** over time
- Ideally, the size of the analog output signal is **proportional** to the size of the variable being measured (i.e. they have a linear relationship)
- Unfortunately, **nonlinear** relationships are very common
- The output signal is typically either **current** or **voltage**



Digital Sensors



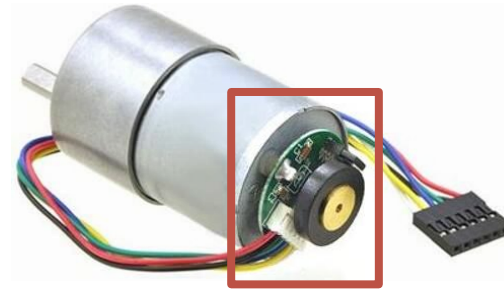
- **Digital sensors** produce an output signal which can only assume a set of **discrete values**
- Often limited to an **on** (1, HIGH, 5V) or **off** (0, LOW, 0V) behavior
- Some sensors measure digital signals while others internally convert analog to digital signals



ME 350 Project Sensors



Infrared Proximity Sensor



Rotary Encoder

What kind of sensors are the rotary encoder and proximity sensors in your ME 350 project?

- A. Both are analog sensors
- B. Both are digital sensors
- C. Proximity sensor is analog and encoder is digital
- D. Proximity sensor is digital and encoder is analog
- E. Not sure...

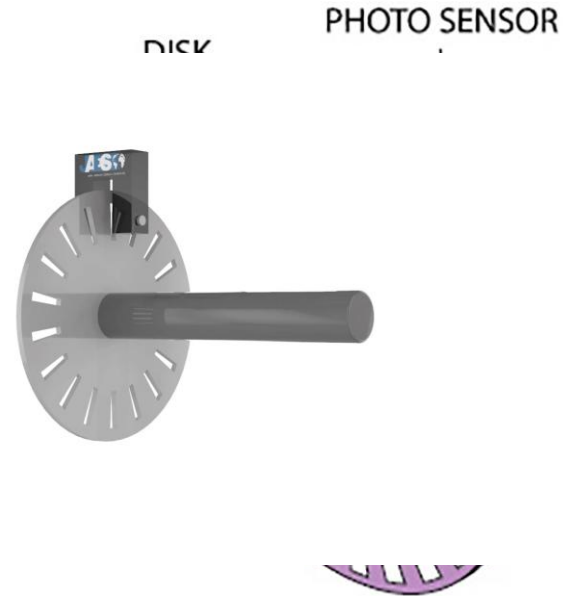
Optical Encoders

- Provide a **digital** output as a result of a linear or angular displacement
- Based on **gating** a light source from a receptor
- Two major categories
 - **Incremental encoders** detect **changes** in rotation or displacement from some position (datum) that can change.
 - **Absolute encoders** give the **actual (or absolute)** angular or linear position relative to a fixed point.



Optical Encoders - Incremental

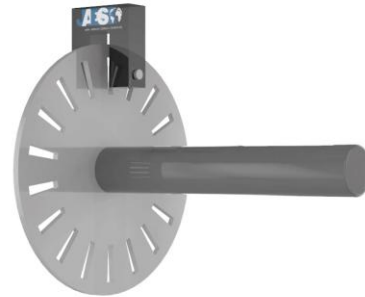
- As the wheel rotates, the photo sensors is turned **on and off**
- By counting these transitions, the rotation of the wheel can be measured



Optical Encoders - Incremental

- As the wheel rotates, the photo sensors is turned **on and off**
- By counting these transitions, the rotation of the wheel can be measured

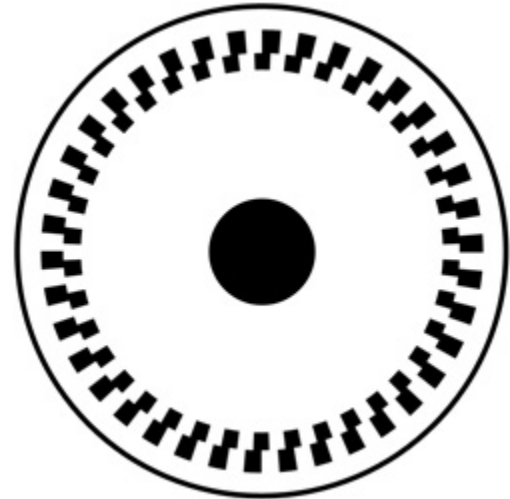
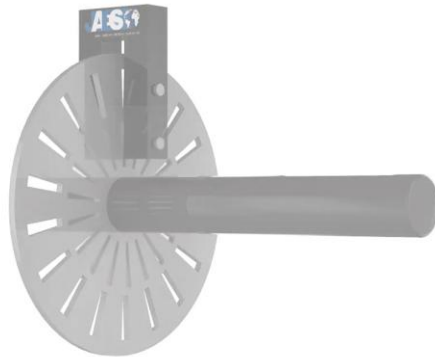
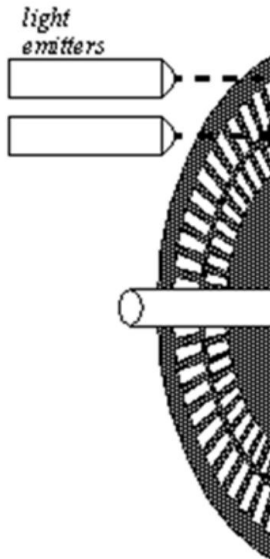
- **NOTE:** This only works if the direction of motion is know...
- How can we figure out the direction?



Optical Encoders – Incremental

How can we resolve the direction of motion?

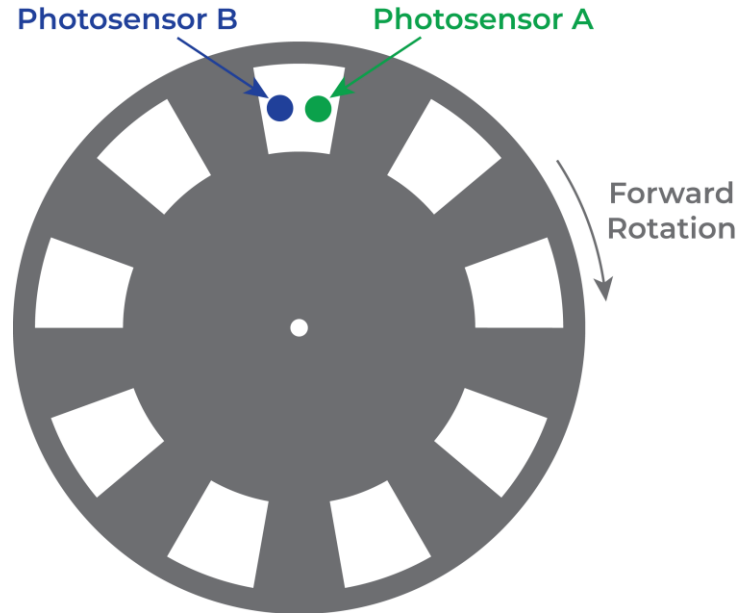
- Use a second LED and photosensor, with a pattern **offset** from the first one
- When the encoder rotates, one channel will **lead** the other allowing for direction to be determined



Optical Encoders – Incremental

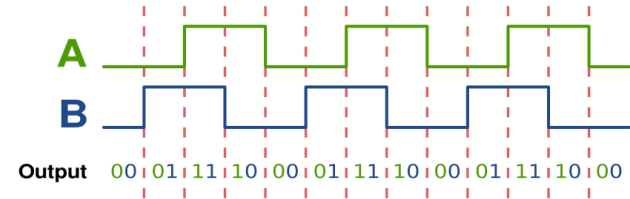
How can we resolve the direction of motion?

- Smarter solution is to use just one grid and mount two sensors with a **half-gate offset**



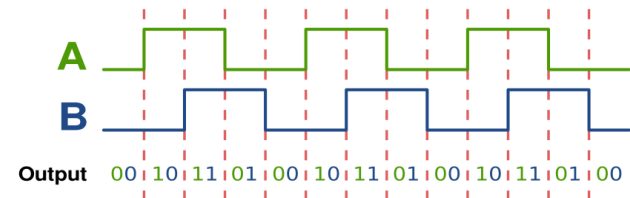
Forward rotation

B "leads" A



Backwards rotation

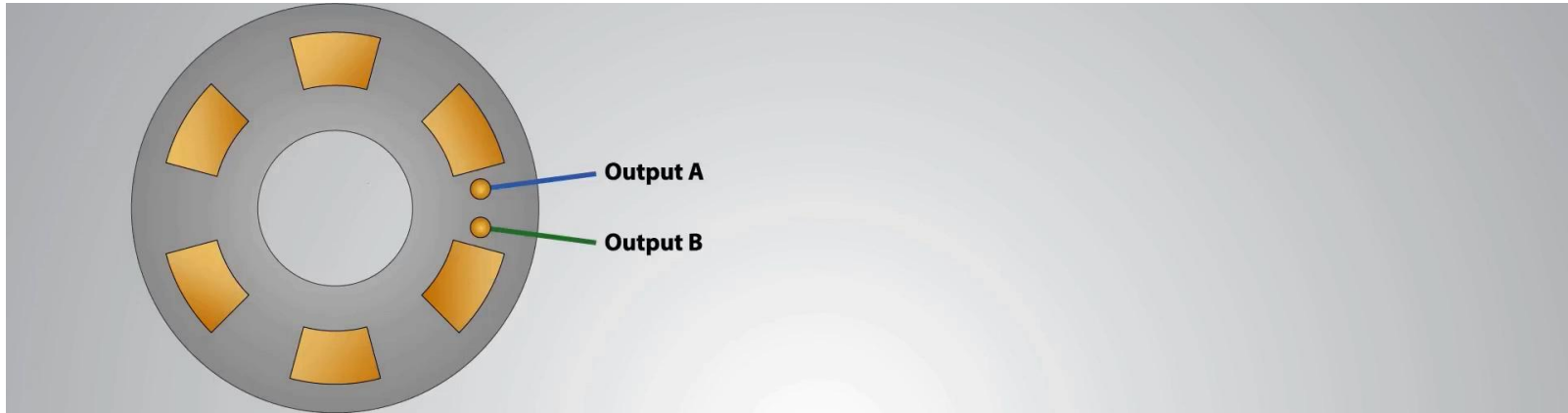
A "leads" B



Optical Encoders – Incremental

How can we resolve the direction of motion?

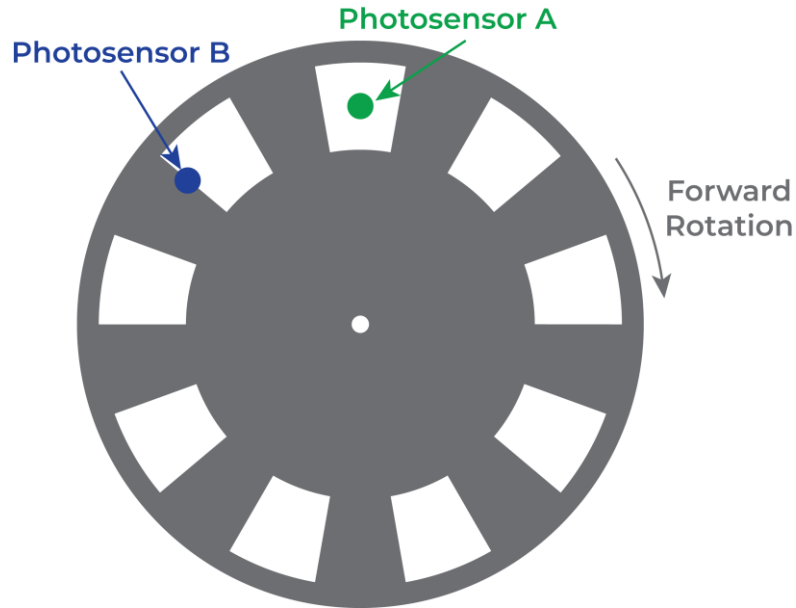
- Smarter solution is to use just one grid and mount two sensors with a **half-gate offset**



Optical Encoders – Incremental

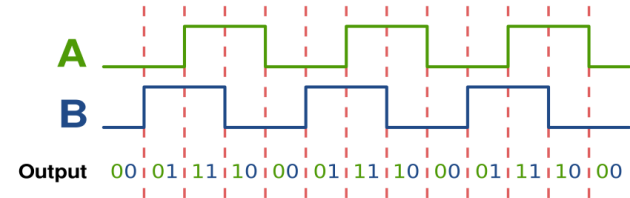
How can we resolve the direction of motion?

- Smarter solution is to use just one grid and mount two sensors with a **half-gate offset**



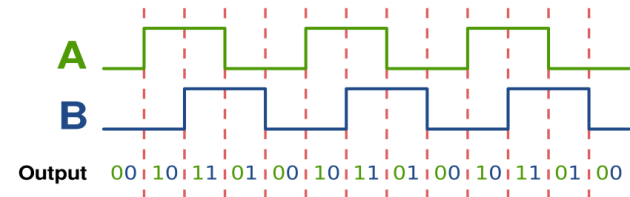
Forward rotation

B "leads" A



Backwards rotation

A "leads" B

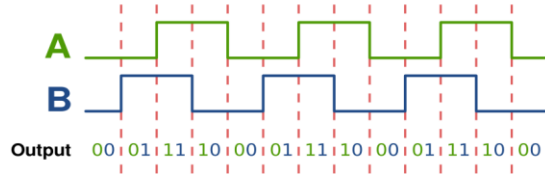


Interpreting an Incremental Encoder Output

Listed in the table is **ALL** possible logic scenarios. Fill out the truth table with 0 (stationary), -1 (backward rotation), 1 (forward rotation), or e (error)

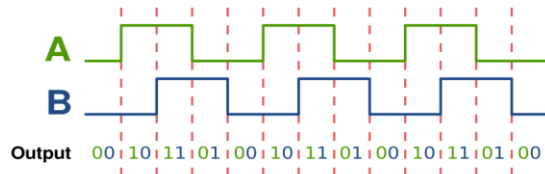
Forward rotation

B "leads" A



Backwards rotation

A "leads" B



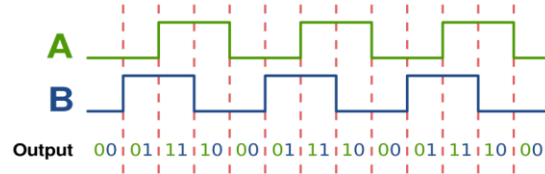
Row #	A Old	B Old	A	B	Action (-1, 0, +1, e)
A	0	0	0	0	
B	0	0	0	1	
C	0	0	1	0	
D	0	0	1	1	
E	0	1	0	0	
F	0	1	0	1	
G	0	1	1	0	
H	0	1	1	1	
I	1	0	0	0	
J	1	0	0	1	
K	1	0	1	0	
L	1	0	1	1	
M	1	1	0	0	
N	1	1	0	1	
O	1	1	1	0	
P	1	1	1	1	

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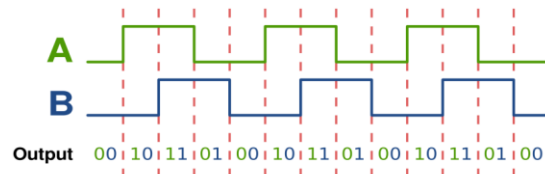
Forward rotation

B "leads" A



Backwards rotation

A "leads" B



Row #	A Old	B Old	A	B	Action (-1, 0, +1, e)
A	0	0	0	0	0
B	0	0	0	1	
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D	0	0	1	1	
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G	0	1	1	0	
H	0	1	1	1	
I	1	0	0	0	
J	1	0	0	1	
K	1	0	1	0	
L	1	0	1	1	
M	1	1	0	0	
N	1	1	0	1	
O	1	1	1	0	
P	1	1	1	1	

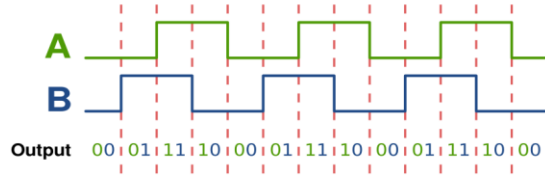
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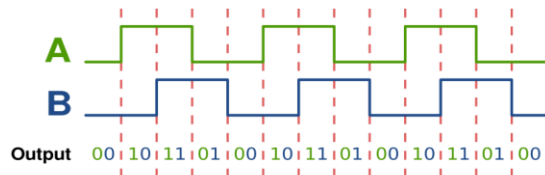
Forward rotation

B "leads" A



Backwards rotation

A "leads" B



Row #	A Old	B Old	A	B	Action (-1, 0, +1, e)
A	0	0	0	0	0
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C	0	0	1	0	
D	0	0	1	1	
E	0	1	0	0	
F	0	1	0	1	
G	0	1	1	0	
H	0	1	1	1	
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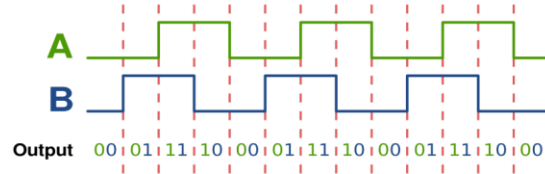
Interpreting an Incremental Encoder Output



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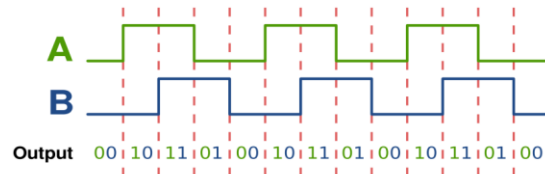
Forward rotation

B "leads" A



Backwards rotation

A "leads" B



Row #	A Old	B Old	A	B	Action (-1, 0, +1, e)
A	0	0	0	0	0
B	0	0	0	1	1
C	0	0	1	0	-1
D	0	0	1	1	
E	0	1	0	0	
F	0	1	0	1	
G	0	1	1	0	
H	0	1	1	1	
I	1	0	0	0	
J	1	0	0	1	
K	1	0	1	0	
L	1	0	1	1	
M	1	1	0	0	
N	1	1	0	1	
O	1	1	1	0	
P	1	1	1	1	

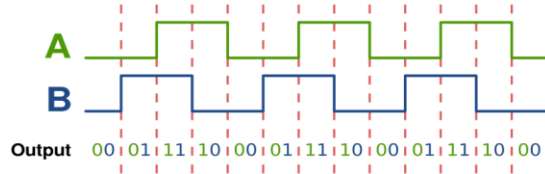
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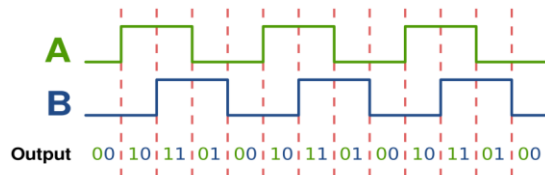
Forward rotation

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Backwards rotation

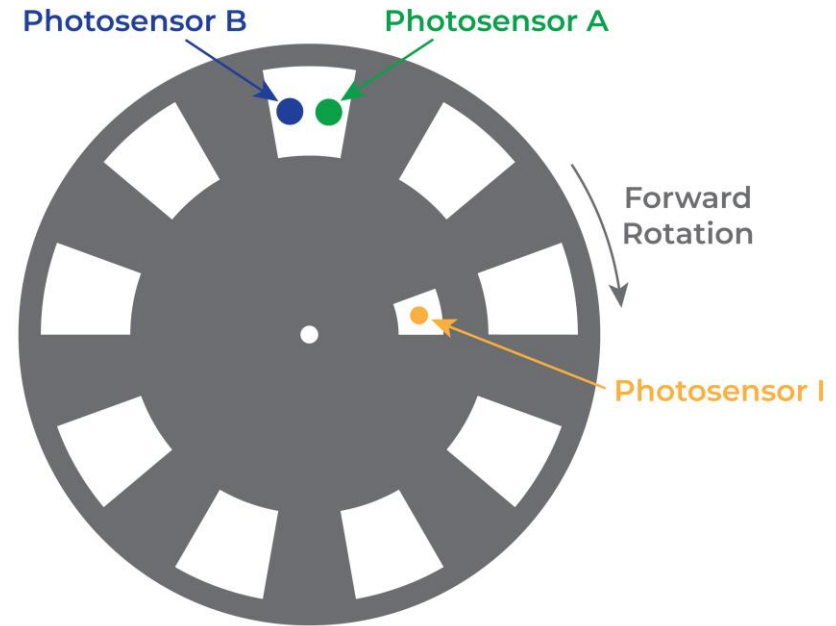
A "leads" B



Row #	A Old	B Old	A	B	Action (-1, 0, +1, e)
A	0	0	0	0	0
B	0	0	0	1	1
C	0	0	1	0	-1
D	0	0	1	1	e
E	0	1	0	0	-1
F	0	1	0	1	0
G	0	1	1	0	e
H	0	1	1	1	1
I	1	0	0	0	1
J	1	0	0	1	e
K	1	0	1	0	0
L	1	0	1	1	-1
M	1	1	0	0	e
N	1	1	0	1	-1
O	1	1	1	0	1
P	1	1	1	1	0

Index (Reference) Pulse

- With an another slit on the disk and another photosensor you can create an **index (reference) pulse**
 - Often called the I-channel or Z-channel of the encoder
- The index count determines the **number of full rotations**
- Also provides a **reference point** for absolute position measurements



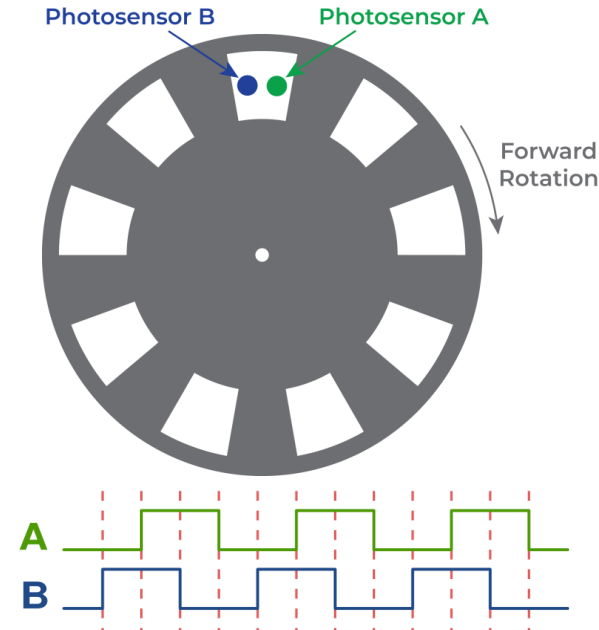
Resolution of an Incremental Encoder

- Counting the **rising edges** of just one channel, you get the **base resolution**,

$$r = 360^\circ/n$$

where n is the number of slots in the disk. Base resolution can also be measured as n counts per revolution (CPR).

- Counting the **rising edge** AND **falling edge** of one channel, you get 2x the base resolution (i.e. $r/2$)
 - Assuming that the gates and blocks of the disk are same width
- Counting the rising edge AND falling edge of **both** channels, you get 4x the base resolution (i.e. $r/4$). This is called **quadrature** decoding
 - Assumes the same as above plus the second photosensor is offset from the first one by half the width of a gate



How to Read an Encoder Datasheet



Properties of the HP HEDS -5505-A14 Incremental Optical Encoder

Specification	Values
Supply Voltage	5 Volts
Supply Current	40 mA
Base Resolution	500 pulses per revolution
Resolution with quadrature	2000 pulses per revolution
Number of Channels	2
Index	none
Maximum Velocity	30000 RPM
Maximum Acceleration	250000 rad/sec ²

- Why are maximum velocity and acceleration values listed?
 - Mechanical: bearing mechanical limits, shock and vibration
 - Electrical: maximum counting speed of the electronics

How to Read an Encoder Datasheet



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Maximum Velocity	30000 RPM
Maximum Acceleration	250000 rad/sec ²

For the provided encoder, calculate the pulse counting frequency at maximum velocity with quadrature decoding

$$f_{\max} = \frac{30,000 \text{ rev}}{\text{min}} \left(\frac{1 \text{ min}}{60 \text{ sec}} \right) (500 \frac{\text{counts}}{\text{rev}} \times 4) = 1 \text{ MHz}$$

Arduino is $\sim 8\text{-}9 \text{ kHz}$

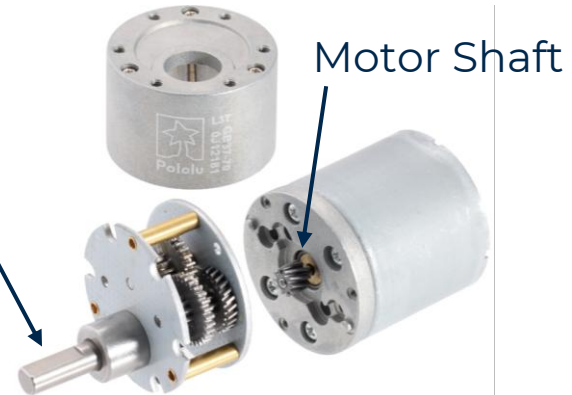
Encoder Used in Your ME 350 Project

- Gearmotor in your project contains a hall effect encoder ***attached to the motor shaft***
- Hall effect encoders are similar to optical encoders but are based on the **hall effect**
 - The hall effect is the principle that a voltage differential is observed in a conductor when placed at the right angle in a magnetic field
- This means that the encoder will read revolutions of the *motor shaft*, **not** revolutions of the *output shaft* of the gearbox



Encoder

Output Shaft



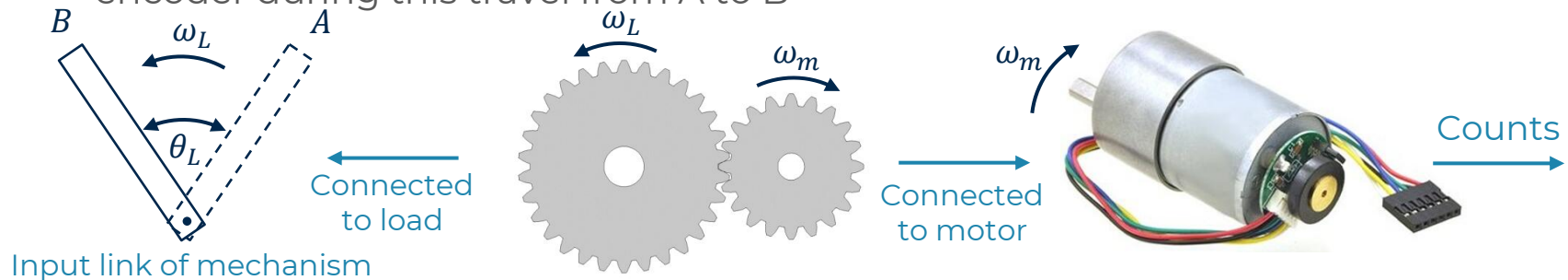
Example

Given:

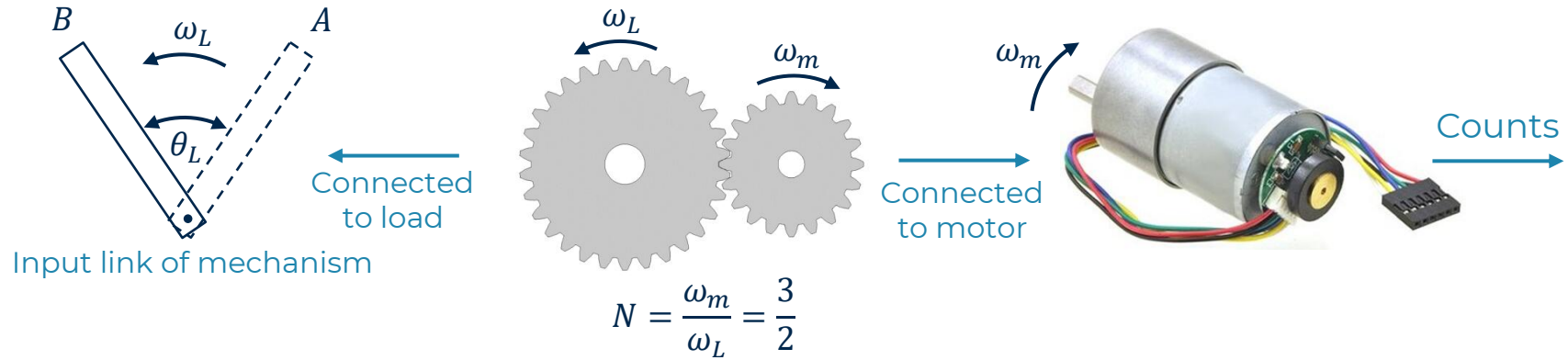
- A gearmotor with 30:1 internal gear ratio, driving the input link of a mechanism through a gear set with 3:2 speed reduction (i.e. $N = \frac{\omega_m}{\omega_L} = \frac{3}{2}$)
- Two-channel encoder, using full quadrature to read the rotation of the motor shaft
- Encoder has a base resolution of 16 counts per revolution (CPR)
- $\theta_L = 158$ degrees of travel of the input link from A to B

Find:

- **The number of counts** (with quadrature decoding) we expect to see from the encoder during this travel from A to B



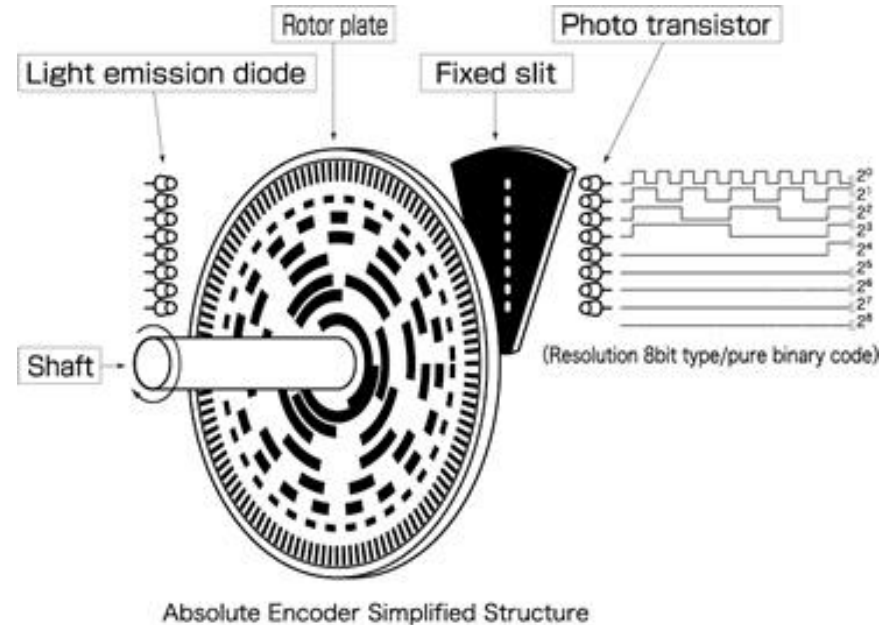
Example



$$\frac{158^\circ}{360^\circ} \left(\frac{3}{2} \right) \left(\frac{30}{16} \right) (16 \times 4) = 1264 \text{ counts}$$

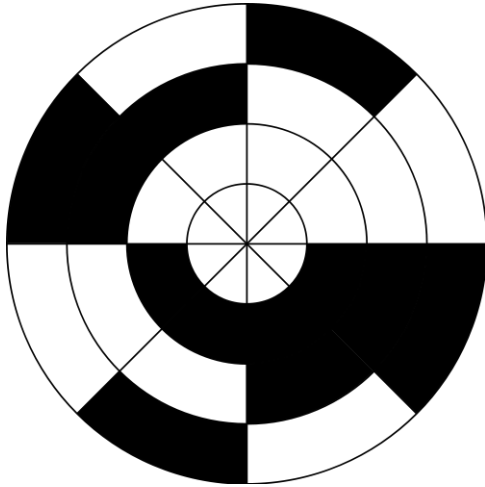
Optical Encoders – Absolute

- Absolute encoders allow direct **digital** readout of **absolute** angular position
- Typically used to measure fractions of a revolution
- Work in a similar way to incremental encoders but the disk has several tracks or rows (call it n rows)
 - Each of these tracks represents a bit in an n bit number



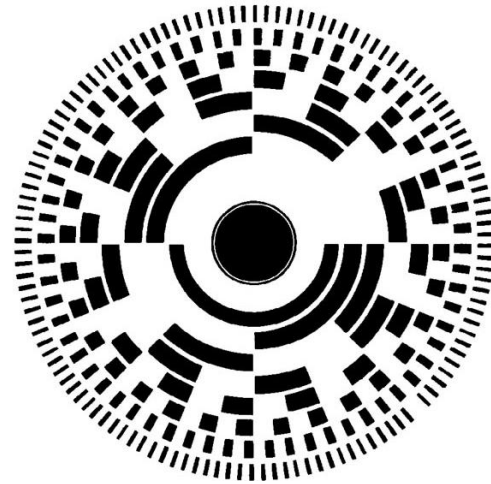
Optical Encoders – Absolute

What is the resolution of each encoder?



3 Bit Encoder

$$\frac{360^\circ}{2^3} = 45^\circ$$



8 Bit Encoder

$$\frac{360^\circ}{2^8} = 1.41^\circ$$

Recap of Binary Numbers



- We count in a base-10 system which we call **decimal**
 - 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

	x 100 (i.e. 10^2)	x 10 (i.e. 10^1)	x 1 (i.e. 10^0)
1	0	0	1
2	0	0	2
...	...	...	...
9	0	0	9
10	0	1	0
11	0	1	1
...	...	...	...
99	0	9	9
100	1	0	0
101	1	0	1

Recap of Binary Numbers



- **Binary** is a base-2 system but the same premise holds
 - 0, 1
 - 4-bit example
 - What is the maximum decimal number we can represent with 4-bits?

$$1111 = 8 + 4 + 2 + 1 = \mathbf{15}$$

	x 8 (i.e. 2^3)	x 4 (i.e. 2^2)	x 2 (i.e. 2^1)	x 1 (i.e. 2^0)
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0

Optical Encoders – Absolute – 4 bit Binary

- Sensors perfectly aligned we have the binary number 0100 which converts to 4
 - Position between 4 and 5

$$\left(\frac{4}{16}\right) \times 360^\circ \text{ to } \left(\frac{5}{16}\right) \times 360^\circ$$

OR

90° to 112.5°

- What if slightly misaligned...
 - Now at 0000 which converts to 0
 - Assumed position between 0 and 1

$$\left(\frac{0}{16}\right) \times 360^\circ \text{ to } \left(\frac{1}{16}\right) \times 360^\circ$$

OR

0° to 22.5°

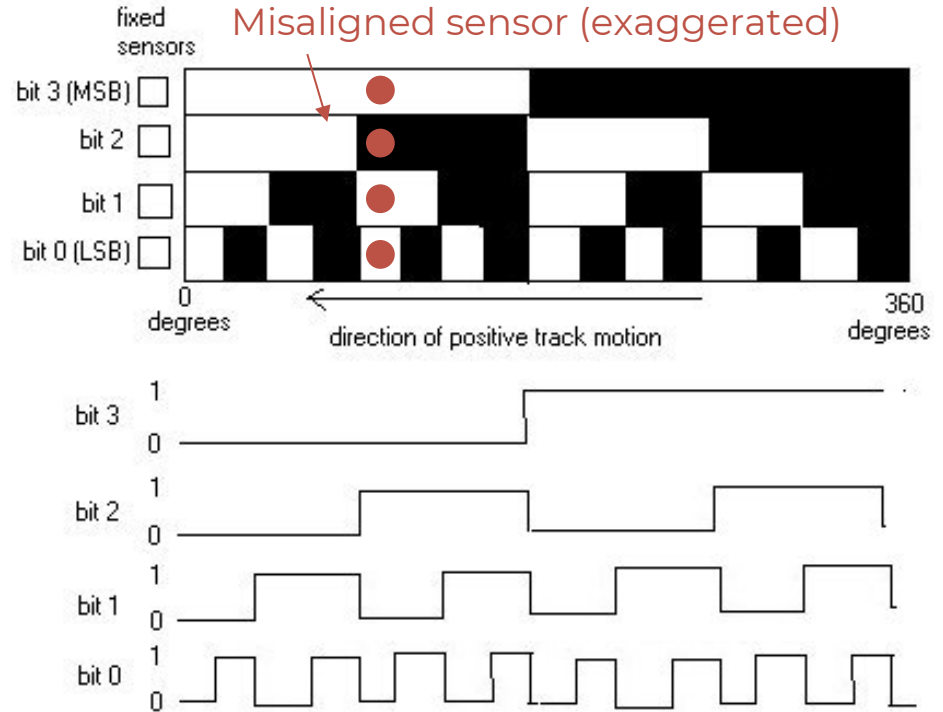
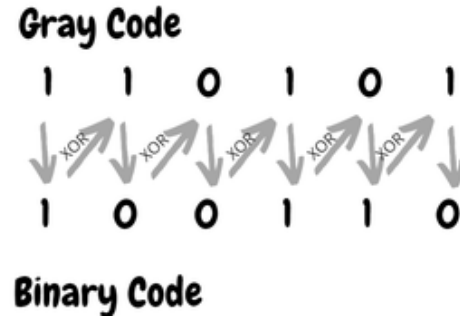
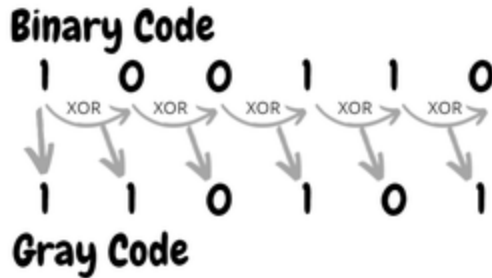


Fig 3 4-Bit binary code absolute encoder disk track patterns

Gray Code

- Known as a reflected binary code, **Gray code** is an ordering of the binary numeral system such that two successive values differ in only one bit
- Convert between Gray and binary using exclusive OR's (don't need to know how to do for this course but interesting to note)



	Gray code			
	4	3	2	1
0	0	0	0	0
1	0	0	0	1
2	0	0	1	1
3	0	0	1	0
4	0	1	1	0
5	0	1	1	1
6	0	1	0	1
7	0	1	0	0
8	1	1	0	0
9	1	1	0	1
10	1	1	1	1
11	1	1	1	0
12	1	0	1	0
13	1	0	1	1
14	1	0	0	1
15	1	0	0	0

Optical Encoders – Absolute – 4 bit Gray

- Gray code is less prone to false counting compared to binary code since only one bit changes at a time
- This improves precision, reliability, and error resistance

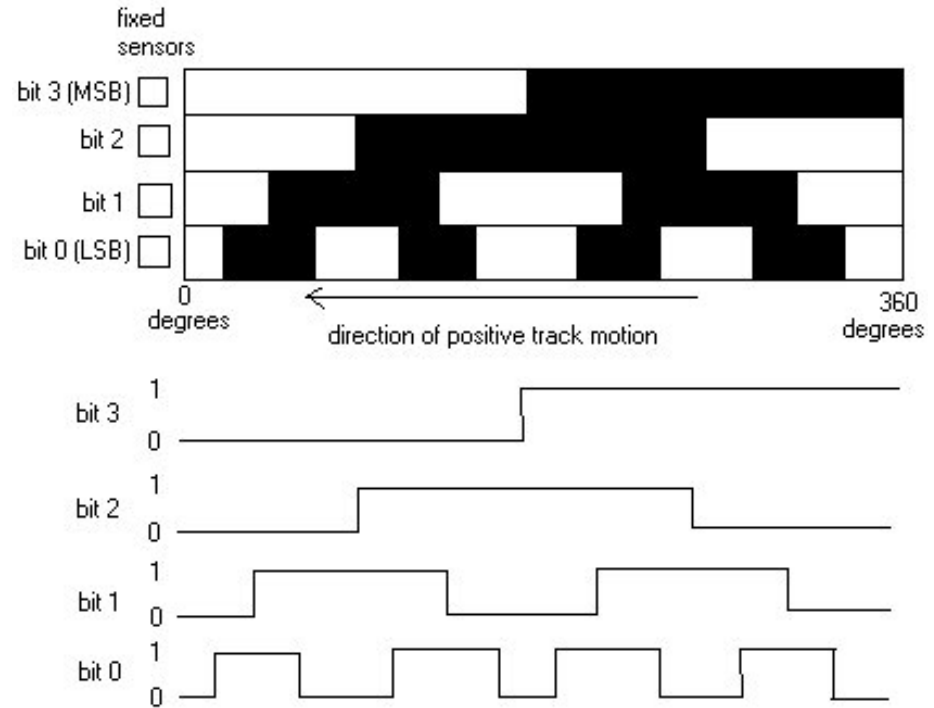
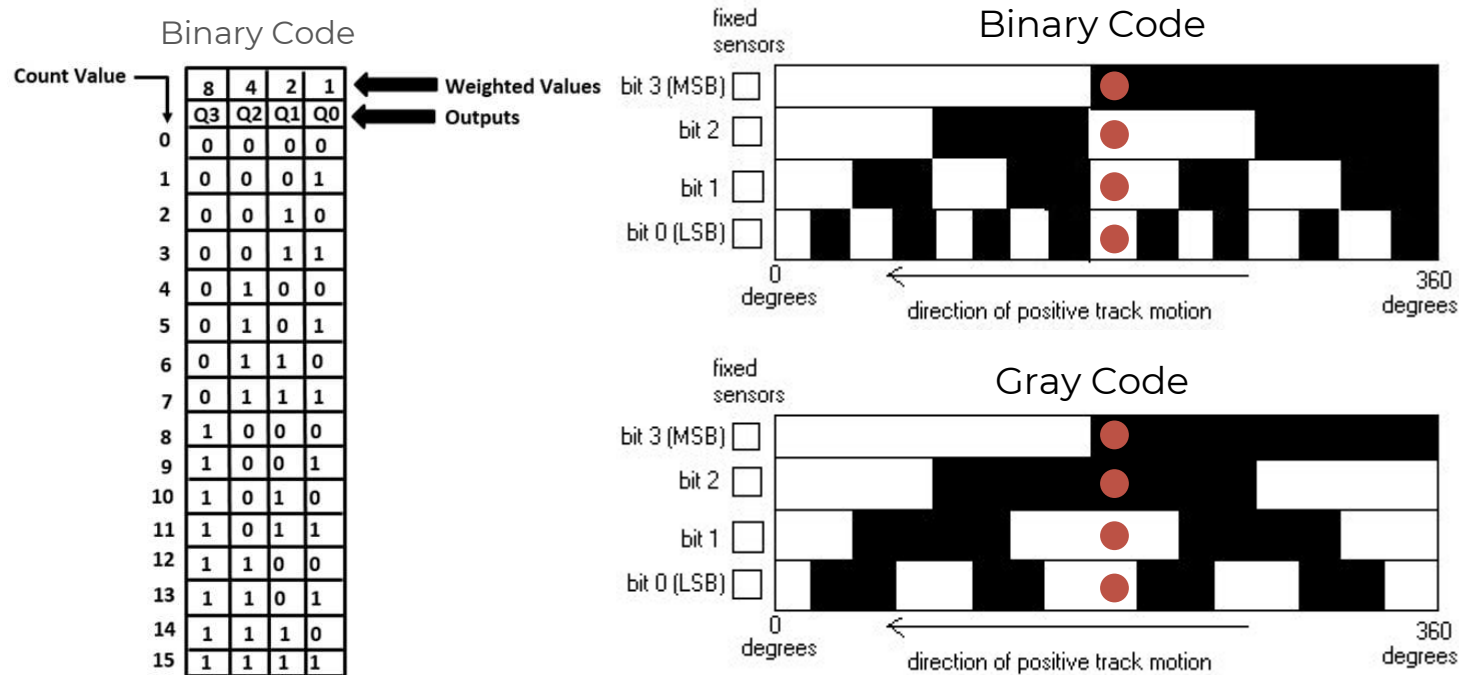


Fig 2. 4-Bit gray code absolute encoder disk track patterns

Binary vs Gray

What is the maximum uncertainty (in degrees) of a 4-bit binary code and a 4-bit gray code absolute encoder if just one sensor is misaligned?



	Gray code			
	4	3	2	1
0	0	0	0	0
1	0	0	0	1
2	0	0	1	1
3	0	0	1	0
4	0	1	1	0
5	0	1	1	1
6	0	1	0	1
7	0	1	0	0
8	1	1	0	0
9	1	1	0	1
10	1	1	1	1
11	1	1	1	0
12	1	0	1	0
13	1	0	1	1
14	1	0	0	1
15	1	0	0	0

Absolute Encoder Codes



Decimal code	Rotation range (deg.)	Binary code	Gray code
0	0-22.5*	0000	0000
1	22.5-45	0001	0001
2	45-67.5	0010	0011
3	67.5-90	0011	0010
4	90-112.5	0100	0110
5	112.5-135	0101	0111
6	135-157.5	0110	0101
7	157.5-180	0111	0100
8	180-202.5	1000	1100
9	202.5-225	1001	1101
10	225-247.5	1010	1111
11	247.5-270	1011	1110
12	270-292.5	1100	1010
13	292.5-315	1101	1011
14	315-337.5	1110	1001
15	337.5-360	1111	1000

- For an absolute encoder, resolution $r = 360/2^n$, where n is the number of bits/tracks
- i.e. 4 tracks gives 2^4 (unique) combinations. Therefore, $r = 360/2^4$ or $r = 22.5^\circ$

Moving Forward...



- Let's look at analog sensors!